UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

THIRD-YEAR FINAL EXAMINATION

03 24347

LH EVOLUTION OF COSMIC STRUCTURE

SEMESTER 2 EXAMINATIONS 2021/22

Time Allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

All symbols have their usual meanings.

A list of useful formulae is provided at the end of the paper.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

Two tables of physical constants and units that may be required will be found at the end of this question paper.

Answer **two** *questions. If you answer more than two questions, credit will only be given for the best two answers.*

1. (a) The amplitude of density fluctuations can be characterized by a power spectrum, $P(k) = \langle|\delta_k|^2\rangle$, where δ_k is the overdensity on the scale of the wavenumber, k. Sketch, on a single plot, the power spectra representative of:

   i. a cold dark matter universe at the end of the radiation-dominated era,

   ii. a hot dark matter universe at the end of the radiation-dominated era.

   Explain the physical causes of their shape and of any differences between the power spectra in (i) and (ii). **[10]**

   (b) Consider a flat universe where the mass density parameter $\Omega_M = 1$ and the Hubble parameter is H. Starting from the equation for the evolution of linear perturbations in an expanding universe,

$$\ddot{\delta} = \frac{3}{2}\Omega_M H^2 \delta - 2H\dot{\delta},$$

   show that the overdensity, δ, grows only linearly with the scale factor, a. Comment on the implications of this finding for the growth of structure in our Universe. **[10]**

   (c) Consider a flat universe with Hubble parameter $H_0 = 70\ \mathrm{km\,s^{-1}Mpc^{-1}}$, mass density parameter $\Omega_{\mathrm{M},0} = 0.3$, and vacuum-energy density parameter $\Omega_{\Lambda,0} = 0.7$. In the Press-Schechter formalism the comoving density, $\frac{dn}{d\log M}$, of halos of mass M is

$$\frac{dn}{d\log M} = \sqrt{\frac{2}{\pi}}\frac{\rho_{\mathrm{M}}}{M}\left|\frac{n_{\mathrm{eff}}+3}{6}\right|\nu e^{-\nu^2/2},$$

   where $\nu = 1.69/\sigma_{\mathrm{M}}$, $\sigma_{\mathrm{M}} \propto M^{-(n_{\mathrm{eff}}+3)/6}$, n_{eff} is the effective spectral index, and ρ_{M} is the comoving matter density. Calculate the comoving density of halos at redshift $z = 0$ with mass of $10^{15}\ \mathrm{M}_\odot$ in a universe where the amplitude of density fluctuations in a sphere with an $8\ \mathrm{Mpc}$ radius (σ_8) is 0.85 and $n_{\mathrm{eff}} = -1.6$. **[10]**

2. (a) List three techniques for finding galaxy groups and clusters. For each technique, explain an advantage and a disadvantage of using that method. **[9]**

(b) In galaxy clusters, the observed relation between X-ray luminosity (L_X) and gas temperature (T) is $L_X \propto T^3$. Assuming the virial theorem holds, show that the expected relation for self-similar clusters at a fixed epoch is $L_X \propto T^2$. If one wanted to explain this difference as arising from a systematic trend in the gas fraction ($f_{\rm gas}$) with temperature, what relation between $f_{\rm gas}$ and T would be required? **[12]**

(c) The two-point correlation function of galaxies $\xi(r)$ is defined such that the expected number of galaxies within a volume V at a distance R from a particular galaxy is $\langle N_g \rangle = n[1 + \xi(r)]V$, where n is the mean number-density of galaxies. Calculate the form of $\xi(r)$ if all galaxies are distributed uniformly on (i) a single sheet or (ii) a filament. Explain qualitatively how the correlation function would be modified if there are $multiple$ sheets or filaments wrapped around voids with characteristic radii R. **[9]**

3. (a) Consider a universe in which the number density of galaxies n depends on luminosity L as

$$\frac{dn}{dL} = \begin{cases} \frac{\Phi_*}{L_*}\left(\frac{L}{L_*}\right)^{\alpha-1} & \text{if } L_{min} \leq L \leq L_*, \\ 0 & \text{otherwise,} \end{cases}$$

where Φ_* = $5.2 \times 10^{-12}\ \mathrm{Mpc}^{-3}$and α = -1.36. The upper and lower luminosity bounds are L_* = $10^{13}\ \mathrm{L}_\odot$ and $L_{min} = 10^6\ \mathrm{L}_\odot$, where $\mathrm{L}_\odot$ is the luminosity of the Sun.

i. Calculate the total number density of galaxies in this universe. **[6]**

ii. Assume 10% of the luminosity in these galaxies is energetic enough to ionize hydrogen and 5% of the ionizing radiation escapes into the intergalactic medium. Calculate whether this universe is reionized if a luminosity of $1.2 \times 10^{29}\ \mathrm{W\ Mpc}^{-3}$ is needed for complete ionization. **[10]**

iii. If the mass-density in this universe is dark-matter dominated, do you expect the dark matter to be cold or hot? Explain your answer. **[4]**

(b) In a hypothetical flat, matter-only universe ($\Omega_M = 1$) that contains only hydrogen, the electron scattering depth from the cosmic microwave background is found to be $\tau_e = 0.09$. Determine the redshift of reionization if the present day Hubble parameter is $H_0 = 70\ \mathrm{km\ s}^{-1}\ \mathrm{Mpc}^{-1}$ and the present day electron density is $n_e = 5.0 \times 10^{-7}\ \mathrm{cm}^{-3}$. Assume that reionization is abrupt.

[10]

Formula Sheet

Friedmann equation
$H^2 = \frac{8}{3}\pi G\rho - \frac{k}{a^2}$

Fluid equation
$\dot{\rho} + 3\frac{\dot{a}}{a}\left(\rho + P\right) = 0$

Acceleration equation
$\frac{\ddot{a}}{a} = -\frac{4}{3}\pi G\left(\rho + 3P\right)$

Equation of State
$P = \omega\rho$

Critical density
$\rho_c = \frac{3H^2}{8\pi G}$

Hubble parameter
$H = \frac{\dot{a}}{a}$

Scale factor (Matter only)
$a(t) \propto t^{2/3}$

Scale factor (Radiation only)
$a(t) \propto t^{1/2}$

Overdensity
$\delta(r,t) = \frac{\rho(r,t) - \bar{\rho}(t)}{\bar{\rho}(t)}$

Dynamical time (gravitational)
$t_{dyn} = \frac{1}{\sqrt{4\pi G\bar{\rho}}}$

Press-Schechter number density
$\frac{dn}{d\log M} = \sqrt{\frac{2}{\pi}}\frac{\rho_{\rm M}}{M}\left|\frac{n_{\rm eff}+3}{6}\right|\nu e^{-\nu^2/2}$

NFW density profile
$\rho(r) = \frac{4\rho_s}{(r/r_s)(1+r/r_s)^2}$

Optical depth
$\tau_e = \int n_e \sigma_T dx$

Feedback energy from supernova
$E_{fb} = \epsilon_{SN}\,\xi\,M_*\,E_{SN}$

Feedback energy from black holes
$E_{AGN} = \epsilon_{BH}\,M_{BH}\,c^2$

Virial theorem
$\mathrm{PE} + 2\,\mathrm{KE} = 0$

Radiation pressure
$P_{rad} = \frac{L}{c}\frac{1}{4\pi R^2}$

Ram pressure condition
$\rho_{icm}v^2 = 2\pi G\Sigma(R)\Sigma_g(R)$

X-ray luminosity
$L_x \propto \rho^2\,V\Lambda(T)$

Correlation function
$\langle N_g\rangle = n[1 + \xi(r)]V$

Astrophysical Constants and Units

Astronomical Unit	AU	1.50×10^{11} m
Parsec	pc	3.1×10^{16} m
Tropical year	y	365.242 mean solar days
Solar luminosity	$L_\odot$	3.84×10^{26} W
Absolute bolometric magnitude of the Sun	$M_{\rm bol}$	$+4^{\rm m}75$
Bolometric correction for the Sun	BC	$-0^{\rm m}08$
Apparent visual magnitude of the Sun	$m_v(\odot)$	$-26^{\rm m}74$
Solar constant	f	1.36×10^{3} W m^{-2}
Solar mass	$M_\odot$	1.989×10^{30} kg
Wien constant	b	2.898×10^{-3} m K
Hubble constant	H_o	70 km s^{-1} Mpc^{-1}
Solar radius	$R_\odot$	6.95×10^{8} m
Distance of the Sun from galactic centre		8.3 kpc
Earth mass	$M_\oplus$	5.972×10^{24} kg
Earth radius	$R_\oplus$	6371 km

Physical Constants and Units

Acceleration due to gravity	g	$9.81\,\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
Ice point	T_{ice}	$273.15\,\mathrm{K}$
Avogadro constant	N_A	$6.022 \times 10^{23}\,\mathrm{mol^{-1}}$
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	$8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$
Stefan constant	σ	$5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\,\mathrm{m^{-1}}$
	$R_\infty hc$	$13.606\,\mathrm{eV}$
Planck constant	h	$6.626 \times 10^{-34}\,\mathrm{J\,s} \equiv 4.136 \times 10^{-15}\,\mathrm{eV\,s}$
$h/2\pi$	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s} \equiv 6.582 \times 10^{-16}\,\mathrm{eV\,s}$
Speed of light *in vacuo*	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
	$\hbar c$	$197.3\,\mathrm{MeV\,fm}$
Charge of proton	e	$1.602 \times 10^{-19}\,\mathrm{C}$
Mass of electron	m_e	$9.109 \times 10^{-31}\,\mathrm{kg}$
Rest energy of electron		$0.511\,\mathrm{MeV}$
Mass of proton	m_p	$1.673 \times 10^{-27}\,\mathrm{kg}$
Rest energy of proton		$938.3\,\mathrm{MeV}$
One atomic mass unit	u	$1.66 \times 10^{-27}\,\mathrm{kg}$
Atomic mass unit energy equivalent		$931.5\,\mathrm{MeV}$
Electric constant	ϵ_0	$8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
Bohr magneton	μ_B	$9.274 \times 10^{-24}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Nuclear magneton	μ_N	$5.051 \times 10^{-27}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	7.297×10^{-3}= 1/137.0
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12}\,\mathrm{m}$
Bohr radius	a_0	$5.2918 \times 10^{-11}\,\mathrm{m}$
angstrom	Å	$10^{-10}\,\mathrm{m}$
barn	b	$10^{-28}\,\mathrm{m^2}$
torr (mm Hg at $0\,^\circ$C)	torr	$133.32\,\mathrm{Pa}$ ($\mathrm{N\,m^{-2}}$)

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.